

MGMT 405 – Managerial Economics

Teaching Note – Module 2: Demand Elasticity and Total Revenue

This note discusses how the own-price elasticity of demand varies along a linear demand curve and then establishes the relationship between demand elasticity and total revenue.

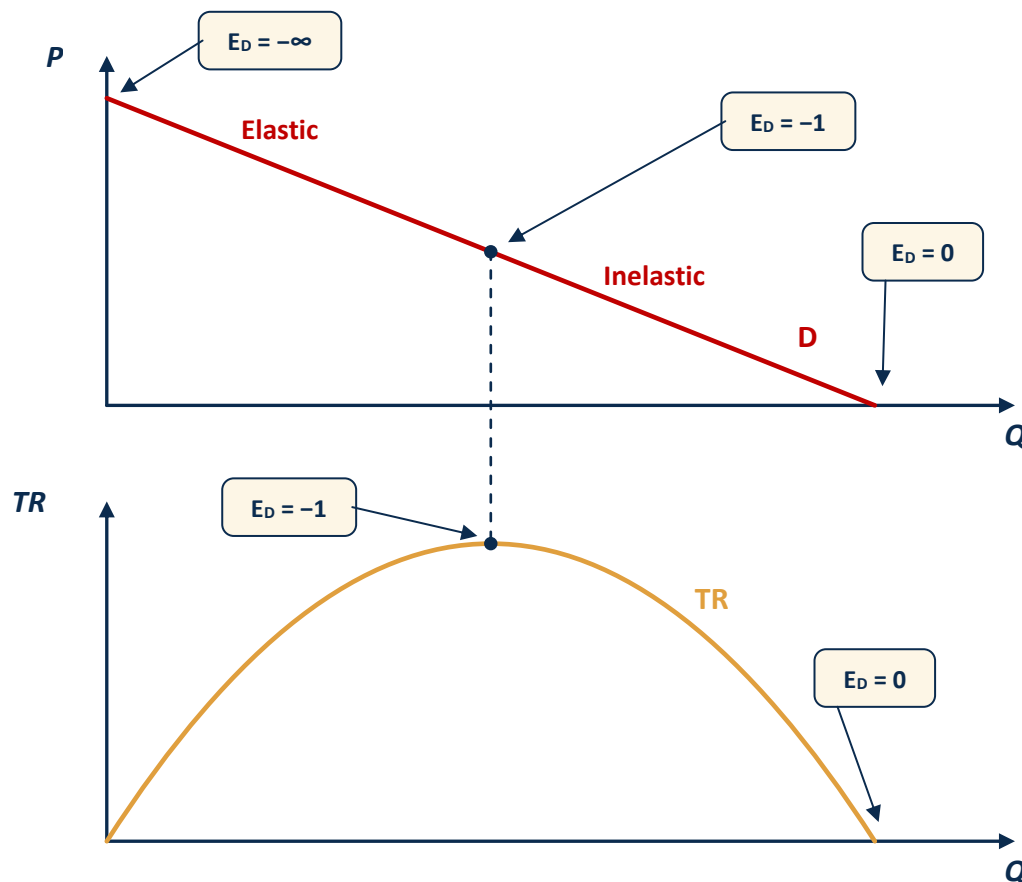
Own-price elasticity of demand along a linear demand curve

The own-price elasticity of demand is defined as follows:

$$E_D = \frac{\% \Delta Q}{\% \Delta P} = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

The first term $\frac{\Delta Q}{\Delta P}$ is the slope of the demand function (Q as a function of P). This slope term is constant in a *linear* demand function. Thus, the second term in the formula, $\frac{P}{Q}$, will determine how the elasticity varies along the curve. We can decompose the curve into three parts:

Recall: we write the demand function as, e.g., $Q = 1,600 - 4P$, but we plot inverse demand, with P on the y-axis and Q on the x-axis.



1. *Elastic*: In the upper left portion of the demand curve, price is very high relative to quantity, making $\frac{P}{Q}$ large. In the elasticity formula, the slope $\frac{\Delta Q}{\Delta P}$ (which is negative) is thus multiplied by a large number, yielding a large negative number. For this reason, the upper left portion of the demand curve is highly elastic.
2. *Inelastic*: In the lower right portion of the demand curve, price is low relative to quantity, making $\frac{P}{Q}$ small. This gives a small negative number for the elasticity – demand is inelastic.
3. *Unit elastic*: Unit elasticity is the point on the demand curve at which the elasticity of demand is equal to -1. At this point, the percentage change in price is exactly offset by the percentage change in quantity. This can be illustrated by writing the elasticity formula in terms of percentage changes:

$$E_D = \frac{\Delta Q/Q}{\Delta P/P} = -1$$

In addition to these three elasticities, we can also compute the elasticity where the demand curve intersects the x and y axes. At the intersection with the y-axis, the quantity is zero, so $\frac{P}{Q} \rightarrow \infty$, and the elasticity is negative infinity. The intuition behind this is that a small decrease in price will yield a change in quantity demanded from zero to a positive value. Even if this change in quantity is small, the *percentage* change is infinite because it is rising from zero.

At the intersection with the x-axis, the price is zero, so $\frac{P}{Q} = 0$, and the elasticity of demand is also zero.

Relationship between elasticity of demand and total revenue

We can now explore the relationship between the elasticity of demand and total revenue. Total revenue is defined as $TR = P \cdot Q$. Starting at the leftmost portion of the graph, when $Q = 0$, we know that $TR = 0$. As the price falls, quantity rises. Note that these changes have opposing effects on total revenue. Since we are on the elastic portion of the demand curve, we know that a small decrease in price causes a large increase in quantity, and total revenue increases. This pattern will hold as price falls through the point of unit elasticity on the demand curve. At this point, total revenue reaches its maximum. This is also the point where marginal revenue is zero, as shown in the Teaching Note on Marginal Revenue. After this point, we enter the inelastic portion of the demand curve. On the inelastic portion of the demand curve, we know that a small decrease in price causes an even smaller increase in quantity, and total revenue falls. Total revenue continues to fall until the demand curve intersects the x-axis, where $P = 0$, and $TR = 0$.